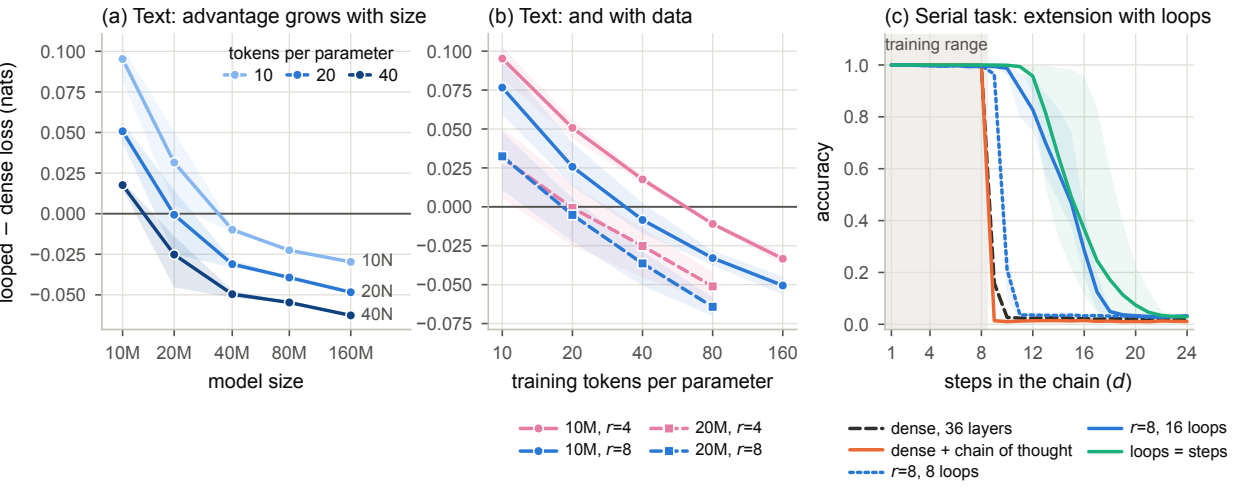
(a) Text: advantage grows with size
looped − dense loss (nats)
tokens per parameter
10
20
40
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
10N
20N
40N
10M
20M
40M
80M
160M
model size
(b) Text: and with data
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
10
20
40
80
160
training tokens per parameter
(c) Serial task: extension with loops
training range
accuracy
1.0
0.8
0.6
0.4
0.2
0.0
1
4
8
12
16
20
24
steps in the chain (d)
10M, r=4
20M, r=4
10M, r=8
20M, r=8
dense, 36 layers
dense + chain of thought
r=8, 8 loops
r=8, 16 loops
loops = steps